When Do Financial Conditions Move Like Monetary Policy?

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Abstract

When can a movement in financial conditions be read as a movement in monetary policy? Two calibrations give an answer. One weights asset prices by how they respond to monetary policy news in announcement windows. The other weights them by their effects on GDP in the FRB/US model. For the combination of asset prices that follows a policy surprise, the two readings are close once they are on a common scale. For other combinations they differ. The model gives the dollar more than twice the weight, relative to equities, that the event study does, and decay assumptions common to both assets do not change that. The difference between the readings is a quarter to a third of the average market move, at horizons from a day to a year. It matters when valuations and the dollar rise together, as in 1998–2001, when the two readings of the same asset movements had opposite signs. It does not when they offset, as in 2022–24, when the readings were the same. Realized unemployment and GDP growth do not separate the two readings.

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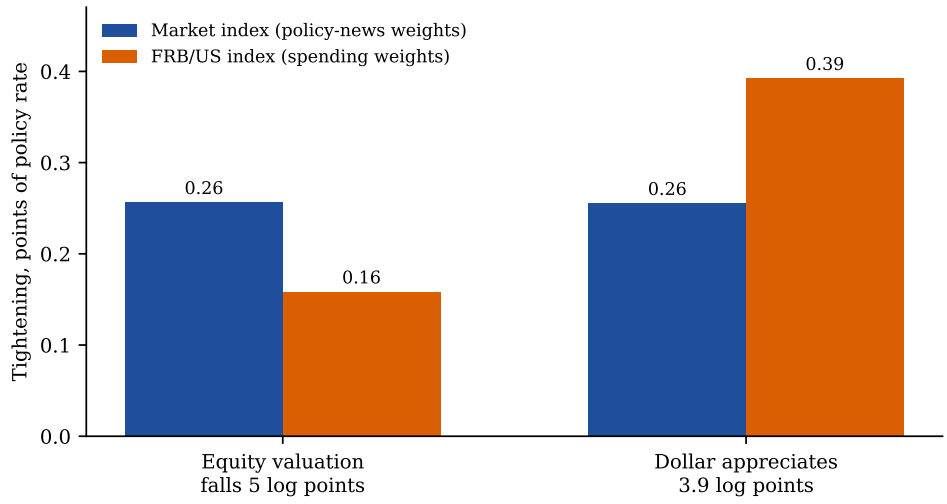


Figure 1: Two Tightenings That Look Alike in Markets

Notes: Market index: points of the two-year yield, with weights from announcement-window responses to policy news. FRB/US index: funds-rate points, the effect on GDP after four quarters of a persistent shift in the variable divided by the effect of a persistent one percentage point rise in the funds rate. The 10-year yield is unchanged in both cases. Table 1 gives intervals.

1 Introduction

Suppose the Federal Reserve leaves the policy rate unchanged, but the dollar appreciates and stock prices fall. Both movements tend to restrain spending. How much tightening has occurred? One way to answer is to ask how large a monetary policy surprise would ordinarily produce a comparable movement in financial markets. Another is to ask how large a change in the policy rate would have a comparable effect on spending. A policymaker might reasonably expect the two answers to be close.

Whether they are close depends on which asset prices have moved. Around a typical monetary policy announcement, long yields, stock prices and the dollar move together in fairly predictable proportions. The two methods give similar readings for that combination. But suppose the tightening comes mainly through the dollar. In the Federal Reserve's model, an appreciation of the dollar has a larger effect on spending than its usual response to policy news would suggest. An equity-led tightening is the converse case. The distinction matters whenever policymakers use the familiar effects of their own announcements to interpret financial developments between meetings.

Figure 1 gives an example. In one case equity valuations fall by 5 log points. In the other the dollar appreciates by 3.9 log points. By the first method, which weights asset prices by how they respond to policy news, each is worth about a quarter of a point of policy tightening (0.256 and 0.255 points of the two-year yield). By the second, which weights them by their effects on GDP in the FRB/US model, the dollar movement is worth 2.48 times the equity movement.

Policymakers read the stance of policy from financial conditions. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher

longer-term bond yields, among other factors" (Powell, 2023), and in March 2024 that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024). A reading of financial conditions is a reading of policy only to the extent that a point of the index means a point of policy. This paper asks when two calibrations of that meaning agree, how large the disagreement is, whether it would have changed the reading of history, and whether later activity favors either one.

I find that the two calibrations agree on the movement that policy itself produces and part ways on the combinations that it does not. On the typical response of long yields, equities and the dollar to a policy surprise, the model prices the response at 0.975 funds-rate points (interval 0.87 to 1.09), against one by construction for the event study. Rescaling the model's weights by that number puts the two readings on one scale. The disagreement lies in composition. The model gives the dollar about 2.49 times the relative weight, against equities, that the event study does, and decay assumptions that are common to both assets do not change that. In the data the difference between the two readings is a quarter to a third of the average market move, at horizons from a day to a year. Whether it matters depends on what moved. In 1998–2001 the two readings of the same asset movements had opposite signs, both small, mostly because of the dollar's appreciation. In 2022–24 they were the same, and both overpredicted the rise in unemployment. Realized unemployment and GDP growth do not separate the two readings in the main sample.

The paper has five parts. First, I define what a policy equivalent is under each method and show why their weights differ (Section 2). Second, I estimate the weights and show where the two readings agree, and how the assumed persistence of asset prices changes the comparison (Section 3). Third, I measure the disagreement on announcements and at horizons from a day to a year (Section 4). Fourth, I examine three historical episodes (Section 5). Finally, I ask whether subsequent activity favors either reading (Section 6). The appendices give the derivations, the construction of the series and the sensitivity of the results.

1.1 Connection to the Literature

Weights and their meaning. Indexes of financial conditions differ mainly in the source of their weights (Hatzius et al., 2010): effects on output from a model (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), principal components (Brave and Butters, 2011), or event-study regressions (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). The older literature on monetary conditions indexes recognized that the weights on asset prices depend on the shock and the dynamics, so that their policy meaning is not invariant (Eika, Ericsson and Nymoen, 1996). I quantify that difficulty with two observed sets of U.S. weights and ask where it matters.

The closest comparisons. Arnaut and Bauer (2024) estimate how an existing index responds to policy surprises, Sack and Swanson (2023) relate FCI-G to policy-induced changes in the two-year yield, and Ajello et al. (2023) translate movements into a growth impulse with model multipliers. Bletzinger, Martorana and Mistak (2026) build a U.S. macro-finance index in units comparable to a

policy rate and compare it with the Goldman Sachs index and FCI-G, finding divergences driven by equities and the strongest tightening in 2022–24. I ask how a policy-news weighting and a macro weighting read the same movements. Boehm and Kroner (2025) show that announcements also move equities and exchange rates along a dimension orthogonal to yield changes, so a movement that departs from the conventional pattern can come from the announcement itself.

Information about activity. Hatzius et al. (2010) build an index conditional on past activity and inflation, Aramonte, Rosen and Schindler (2013) show that comparisons among indexes depend on the crisis in the sample, and Lombardi, Manea and Schrimpf (2025) separate safe yields from priced risk and find that risk predicts activity more strongly. Adrian, Boyarchenko and Giannone (2019) show that conditions shift the lower quantiles of growth, and Brave and Butters (2011) build an index of risk, credit and leverage. Caballero, Caravello and Simsek (2026) define FCI*, the level of financial conditions consistent with output at potential. The neutral rate matters for the level of every index (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020); I measure the inputs relative to r^* and do not estimate a neutral level for the index (Appendix B).

2 Two Ways to Read a Financial Movement

2.1 Two Meanings of a Policy Equivalent

Take a movement in three asset prices: the 10-year Treasury yield, equity valuation and the real broad dollar. Each is measured relative to r^* and to its 1973–2026 average (Appendix B). I call a reading of the movement a policy equivalent if it says how much tightening of the policy rate the movement amounts to. There are two natural ways to compute one.

The market index starts from policy. At each Federal Open Market Committee announcement I measure the policy surprise with the factor of Nakamura and Steinsson (2018), scaled so that one unit raises the two-year yield by one percentage point, and I measure how the three asset prices respond in a narrow window around the announcement. The weights are chosen to recover the size of the surprise from the three asset prices together. A movement is then worth the size of the surprise that would, on average, produce it, in points of the two-year yield. A policy surprise of one unit reads one.

The FRB/US index starts from spending. Each asset price is weighted by its effect on GDP four quarters later in the Federal Reserve's FRB/US model (Brayton, Laubach and Reifschneider, 2014), in the experiment in which the asset price shifts persistently, divided by the effect of a persistent one percentage point rise in the federal funds rate. A movement is then worth the funds-rate change with the same effect on GDP, in funds-rate points.

The first calculation reads a movement through what policy does to markets. The second reads it through what markets do to spending. Both are legitimate. They need not agree.

2.2 Why the Weights Differ

Consider the dollar. It can be a reliable indicator of a policy surprise because it responds consistently when the expected path of U.S. interest rates changes. How useful it is as an indicator depends on how much it also moves for other reasons, and on what yields and equities already reveal. Its weight in a spending index depends on different things: how an appreciation affects net exports, and how long the appreciation lasts. An asset can be an excellent indicator of a policy surprise and have a modest effect on spending. It can also be a noisy indicator and have a large effect on spending. The same is true of equities, whose effect on spending runs through wealth and the cost of capital. A fall in equity prices that reflects a higher discount rate is policy-like, and a fall that reflects weaker earnings prospects is not policy at all. The event study averages over announcements, where the first kind dominates. The model prices an equity movement by its estimated effect on spending, whatever its source. Neither calculation is wrong about its own question.

Two equations make the argument precise. Let x be the movement in the three asset prices around an announcement, s the policy surprise, and ψ the vector of responses of the asset prices to a unit of surprise:

$$x = \psi s + u. \tag{1}$$

The vector u collects the ways in which a particular movement departs from the usual response to conventional rate news. It has covariance Σ. Even an announcement can produce such a departure. The market weights w_M recover s from x by generalized least squares, $w_M \propto \Sigma^{-1}\psi$, normalized so that $w_M'\psi = 1$. The FRB/US weights b_h are the effects of the three asset prices on GDP at horizon h in funds-rate points. The model prices the usual response ψ at $b_h'\psi$, which I report in its own units, and dividing by it gives the normalized weights $w_{B,h} = b_h/(b_h'\psi)$, so that both indexes read one on a conventional policy surprise. The difference between the two readings of a movement x is then

$$D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u, \tag{2}$$

because the policy part ψs cancels. In words, the two methods disagree only to the extent that asset prices move in unusual proportions and the methods value those departures differently. An unusual movement is not enough by itself: if the weights are the same, there is nothing to value differently. A difference in weights is not enough either: if the asset prices keep moving together as they do around policy news, the difference has nothing to act on. That is why two indexes can track each other closely for decades and still give different readings during an equity boom or a dollar surge. Throughout, D is the difference computed on the movements of the three asset prices over the horizon in question, and in Section 6 it is computed on 12-month changes. Appendix A derives the weights.

2.3 Putting Numbers on It

Table 1 applies both sets of weights to the two movements of Figure 1. The market index reads the two alike, as the weights imply that a 5 log point fall in equity valuation and a 3.9 log point

Table 1: Two Financial Movements that the Market Index Rates Alike

	Market index	90% interval	FRB/US index
Equity valuation falls 5 log points	0.256	[0.147, 0.351]	0.158
Dollar appreciates 3.9 log points	0.255	[0.081, 0.443]	0.392
Dollar effect over equity effect	1.00	[0.33, 2.30]	2.48
Typical joint response to a unit of policy news	1.00		0.975 [0.87, 1.09]

Notes: Weights applied to the stated movements, with no change in the 10-year yield. Market index: two-year-yield points per movement, weights from announcement-window responses (Table 2); the interval takes the 5th and 95th percentiles over bootstrap samples of the announcements. FRB/US index: funds-rate points; the effect on GDP after four quarters of a persistent shift in the variable, divided by the effect of a persistent one percentage point rise in the funds rate. Typical joint response: the index applied to the responses of the three variables to a unit of policy news (one by construction for the market index).

appreciation of the dollar are each about a quarter of a policy move. The FRB/US index gives the dollar movement 2.48 times the effect of the equity movement. The ratio of ratios, the model's dollar-to-equity ratio over the market's, is 2.49 (interval 1.02 to 6.81). Neither movement is a policy surprise, so nothing in the construction requires the readings to agree. Whether the difference is large in practice depends on how often asset prices move in such proportions, which is the question of Section 4.

3 Why the Two Answers Agree on a Policy Move

3.1 The Estimated Weights

Table 2 gives the market weights. A policy move that raises the two-year yield by one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and raises the real dollar by 4.22 percent. The data are the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): 253 announcements from January 1996 to September 2026. Each variable carries a substantial part of the policy move, with shares of 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar. The weights are imprecise, the weight on the 10-year yield most of all (Appendix C.1).

The weights say what the variables do when policy moves, not what causes their movements at other times. A rise in stock prices that comes from news about earnings has little to do with policy, and the index treats it as if it did. An announcement can also reveal the Fed's view of the economy, and then the responses mix policy and information (Nakamura and Steinsson, 2018). I measure the responses as markets receive them and set the information effect aside. The index measures policy-equivalent tightening. It is not a forecast of activity.

Table 3 compares the market weights with the FRB/US weights. The two sets agree on the 10-year yield, where the FRB/US weight is 0.94 times the market weight. The model puts less weight on equity valuation (0.62 times) and more on the dollar (1.54 times). On a policy move the difference does not matter, because the policy part cancels in equation (2). On a departure from

Table 2: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.128, 1.126]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.070, −0.029]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.114]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 500 bootstrap samples of the announcements. Share: weight times response.

Table 3: FRB/US and Market Weights on the Three Common Variables

	FRB/US weight	Market weight	90% interval	FRB/US response	Market response
10-year yield (pp)	0.569	0.609	[0.128, 1.126]	0.37	0.53
Equity valuation (percent)	−0.032	−0.051	[−0.070, −0.029]	−13.18	−7.78
Real dollar (percent)	0.100	0.065	[0.021, 0.114]	1.62	4.22

Notes: FRB/US weight: effect on GDP after four quarters of a persistent unit shift in the variable, divided by the effect of a persistent 1 percentage point rise in the federal funds rate (0.4525 percent of GDP); funds-rate points per unit. Market weight: two-year-yield points per unit (Table 2); the interval is its bootstrap interval. FRB/US response: change in the variable four quarters after a persistent 1 percentage point funds-rate rise (VAR expectations). Market response: change in the announcement window per unit of policy news.

the usual response, the model gives a fall in equity prices its estimated effect on output, and the market index treats it as policy-equivalent.

3.2 The FRB/US Reading of a Policy Move

The model prices the usual response of the three variables to a unit of policy news at 0.925, 0.975 and 0.930 funds-rate points at two, four and eight quarters (Table 4). At four quarters the 90 percent interval is 0.87 to 1.09. The market index prices the same response at exactly one. A value near one is informative because nothing in the construction of the FRB/US weights refers to policy announcements. They are the effects on GDP of persistent shifts in each variable. Had the usual response to a policy surprise been mostly a dollar movement, or mostly an equity movement, the model could have priced it far above or below one. That it prices it near one means that the event study and the model describe the same policy move with about the same strength. This is the empirical content of agreement: two calibrations built for different purposes, one from announcement windows and one from a macroeconomic model, assign nearly the same number of funds-rate points to the same movement in asset prices. The two do not differ significantly at any horizon.

Dividing the model's weights by 0.975 puts both indexes on one scale and makes them agree exactly on a policy move. That is a normalization and not a finding. The finding is the number before it was divided out.

Table 4: Agreement on Announcements and the Contrast in Composition

FRB/US horizon	Joint response to policy news [90%]	Normalized equity weight	Normalized dollar weight	Dollar over equity, FRB/US	Ratio of ratios [90%]
2 quarters	0.925 [0.83, 1.03]	−0.032	0.089	2.15	2.17 [0.89, 5.93]
4 quarters	0.975 [0.87, 1.09]	−0.032	0.103	2.48	2.49 [1.02, 6.81]
8 quarters	0.930 [0.83, 1.04]	−0.033	0.106	2.53	2.54 [1.04, 6.96]

Notes: FRB/US weights for GDP at the horizon shown, persistent-shift experiment, in funds-rate points; the joint response is the FRB/US weights applied to the responses to a unit of policy news (market index: one). Normalized weights: FRB/US weights divided by that response, so that both indexes read one on a conventional surprise; market weights are 0.609 (10-year yield), −0.051 (equity valuation) and 0.065 (dollar). Ratio of ratios: the FRB/US dollar-over-equity ratio divided by the market's (1.00); the interval comes from the bootstrap of the market weights, and the FRB/US weights are held fixed.

3.3 Composition or Persistence?

The comparison confounds two things. A persistent dollar appreciation in FRB/US and the dollar's response to policy news are different experiments, so a difference in weights could reflect the initial composition of the movement or the path that follows it. Figure 2 separates them with the same set of calculations under common decaying paths for the three variables, with half-lives of 8, 4 and 2 quarters.

Persistence changes how much tightening a policy move represents. The model's reading of the usual response falls from 0.975 with persistent paths to 0.83 (interval 0.74 to 0.93) with a half-life of four quarters and 0.72 with a half-life of two. Persistence does not change which movement counts more. With the same decay for the dollar and for equities, the model's dollar-to-equity ratio is 2.50, 2.52 and 2.55 at half-lives of 8, 4 and 2 quarters, against 1.00 for the market weights (interval 0.33 to 2.30). Different half-lives for the two assets give a wider range, 1.88 to 3.36, and none near one. The contrast between the two weightings is a contrast in the effect on GDP per unit of each asset price.

4 How Much Do the Two Readings Disagree in Practice?

The disagreement depends on how often asset prices move in proportions other than the usual response. I measure it as the mean absolute value of D divided by the mean absolute value of the market move, over the same windows. A ratio of 0.3 means that the typical difference between the two readings is a third of the typical move.

On the 253 announcements, with the weights re-estimated without the event being evaluated, the ratio in intraday windows is 0.19 (interval 0.09 to 0.36), and the 90th percentile of the difference is 0.023 against a mean absolute market move of 0.053. These are the windows in which the weights are estimated, with each event held out.

The ratio is larger in daily data and at longer horizons. On daily changes of the same three assets, 1996–2026, it is 0.29 on announcement days and 0.31 on other days; at one week it is 0.26 and 0.28, and at 22 trading days 0.26 and 0.25 (Table 5). No difference between announcement and other windows is distinguishable from zero. Announcement days contain a great deal of financial

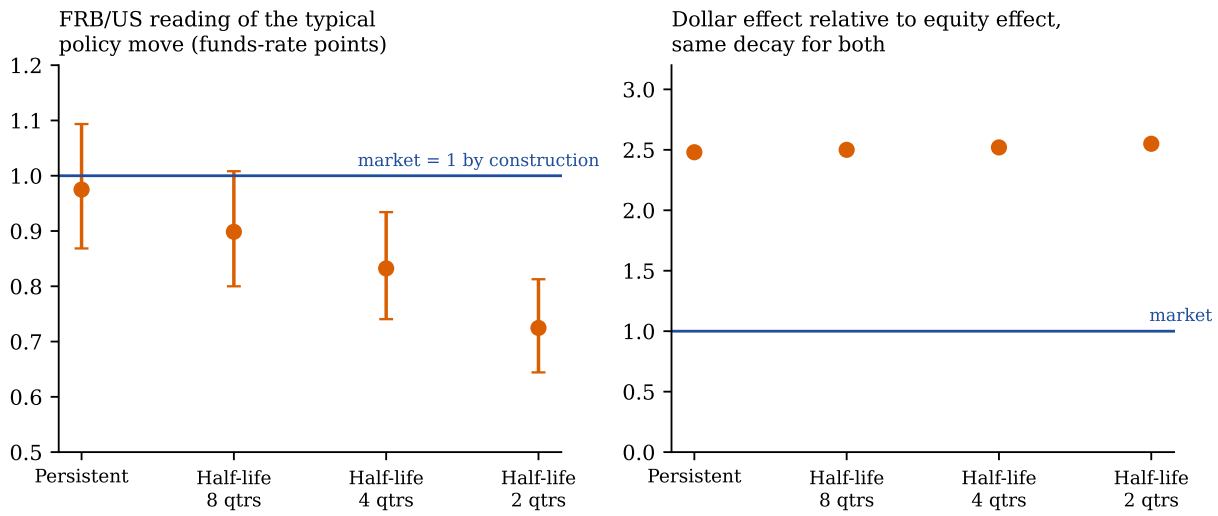


Figure 2: Persistence Changes the Scale, Not the Contrast

Notes: Left: the FRB/US weights at four quarters applied to the responses of the three variables to a unit of policy news, with the variables assumed to persist or to decay with the half-life shown; the bars are 90 percent intervals from bootstrap samples of the announcements. Right: the model's dollar effect over its equity effect, with the same decay for both; the line is the market index's ratio. The persistence of the variables' levels is a different object from the persistence of their responses to policy news; the half-lives of the levels since 1996 are 20 to 26 quarters, and the equity response to policy news does not decay within 20 trading days.

news besides the announcement. The central economic point is that the usual response is special: whole announcement days look like other days.

For windows that end on any day, the ratio is 0.32, 0.28, 0.26, 0.30 and 0.33 at 1, 5, 22, 66 and 252 trading days. A window of 22 or more days contains announcements in any case; removing the moves on announcement days gives 0.32, 0.29, 0.28, 0.31 and 0.34, within 0.02 of those. In 12-month changes of the monthly series, the ratio is 0.33 (interval 0.16 to 0.66), and the difference exceeds 0.25 in 36 percent of months (Figure 3). The months overlap, so runs in this series are not independent evidence of persistence. The statistic is a sample summary with wide intervals. A quarter to a third is a fair description across horizons from a day to a year.

The intraday and daily figures do not isolate a source for the gap between 0.19 and 0.3. The intraday windows are narrow and the weights exclude the event evaluated. The daily comparison uses nominal assets and weights fitted on the same announcements (Appendix D). The comparison across horizons is the cleaner result: the disagreement does not shrink as the window lengthens.

5 Would a Policymaker Have Read History Differently?

Three episodes show when the choice of weights matters. For each I report two comparisons that are easily confused. The first is the difference between the full FRB/US index and the market index in average level, which includes variables the market index does not have and the effect of centering

Table 5: The Difference Between the Readings on Announcement Days and Other Days

Horizon	Announcement windows [90%]	n	Other windows [90%]	n
One day	0.29 [0.16, 0.56]	251	0.31 [0.17, 0.62]	7339
One week	0.26 [0.14, 0.56]	251	0.28 [0.15, 0.59]	1265
22 trading days	0.26 [0.15, 0.56]	248	0.25 [0.13, 0.52]	98

Notes: Mean absolute difference D between the normalized FRB/US reading and the market reading, as a ratio to the mean absolute market move, on the same three assets, 1996–2026 (daily 10-year yield, nominal broad dollar and CAPE; the nominal dollar and no adjustment for r^* stand in for the real dollar and the r^* transformation). Announcement windows end on a scheduled FOMC announcement date; other windows are non-overlapping and contain none. Intervals from joint bootstrap samples of the windows and of the announcements that set the weights.

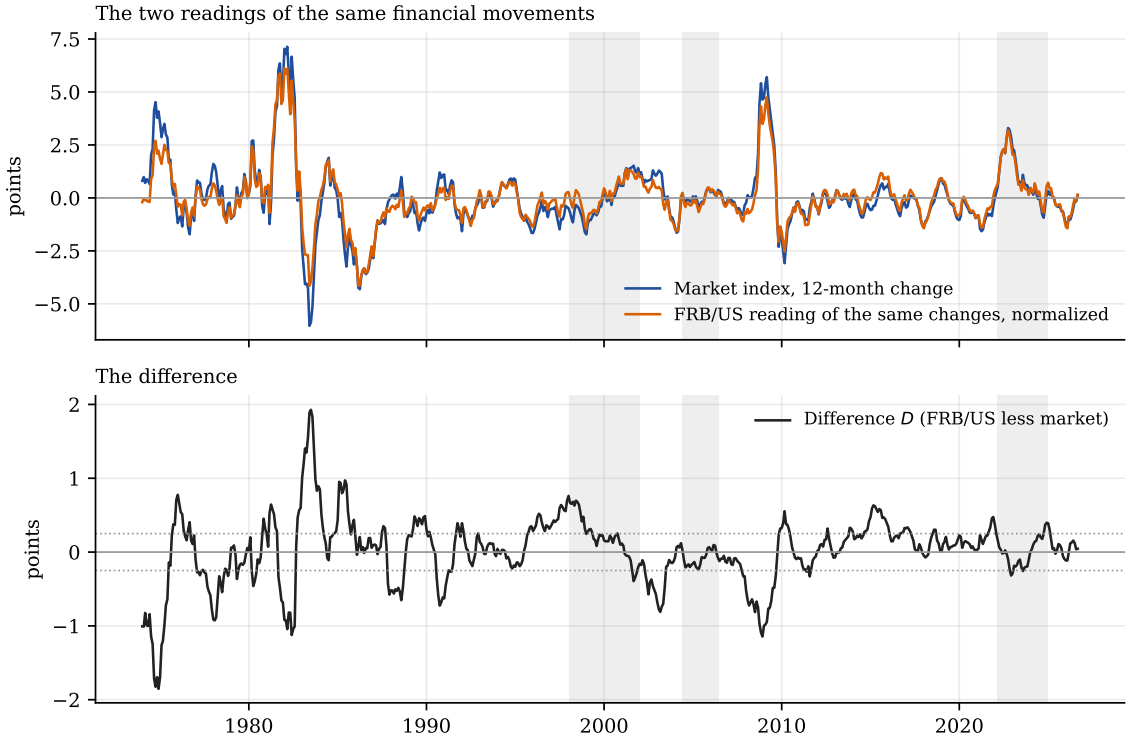


Figure 3: Two Readings of the Same Financial Movements

Notes: Monthly, 12-month changes of the equity valuation, the 10-year yield and the dollar relative to r^*: the 10-year yield less ten-year expected inflation less r^*, the earnings-yield gap to r^* converted to CAPE log points, and the real dollar (Appendix B). Top: the change in the market index and the FRB/US reading of the same changes, normalized so that a conventional policy surprise reads one. Bottom: the difference D, FRB/US less market, with dotted lines at ±0.25. Shaded: 1998–2001, 2004–06 and 2022–24.

Table 6: Two Readings of Three Episodes

	1998–2001	2004–06	2022–24
Full indexes, average level (points)			
FRB/US index less market index [90% interval]	1.73 [0.27, 3.03]	−0.17 [−0.81, 0.41]	0.96 [−0.05, 1.79]
of which: equity weights	1.29	0.65	0.53
Same three assets, 12-month change (points)			
Market reading	−0.06	0.04	1.02
FRB/US reading	0.16	−0.05	1.01
Difference D	0.22	−0.09	−0.01
Change in the unemployment gap over the next 12 months (points)			
Realized	0.35	−0.41	0.27
Forecast from the market reading	0.67	0.13	0.65
Forecast from the FRB/US reading	0.78	0.10	0.66

Notes: Top block: difference in the average levels of the full FRB/US index (with house prices) and the market index relative to r^*, in points, and the part that comes from the equity weights (Table C.1). Middle block: 12-month changes of the market index and of the normalized FRB/US reading, computed on the same three assets and averaged over the episode; D is their difference. Bottom block: change in the unemployment gap over the following 12 months, averaged over the episode's origins, and recursive out-of-sample forecasts of it from the market reading and from the FRB/US reading (own lags included). A gap in levels can coexist with agreement in changes.

each variable on its average. The second applies the two weightings to the same three assets in 12-month changes, which isolates the weights. A gap in levels can coexist with agreement in changes. Table 6 gives both, and Figure 4 shows the matched readings.

1998–2001. The market index read the 12-month changes of these years as slightly easing, at −0.06 points on average. The FRB/US reading of the same asset movements was slightly tightening, at 0.16. The difference of 0.22 is mostly the dollar. The dollar appreciated by 4.18 log points a year on average, which adds 0.16 to D because the model gives the dollar more weight; equity valuations rose by 2.70 log points a year, which adds 0.05. Someone following the market index would have concluded that financial conditions were easing modestly, and someone following the FRB/US reading that they were tightening modestly. Both are small compared with 2022–24. Unemployment rose by 0.35 points over the following year, against forecasts of 0.67 from the market reading and 0.78 from the FRB/US reading.

The full indexes differ much more in levels. The FRB/US index read 1.73 points tighter than the market index on average (interval 0.27 to 3.03), and 1.29 of that difference comes from the equity weights. Valuations were far above their average, and the market index, which gives each point of valuation more weight, read conditions as easier. The level and the change are different objects. The level difference is the weights applied to a persistent departure of valuations from their average; the difference in annual changes is the weights applied to what moved that year.

2022–24. The two readings were the same. The market reading of the 12-month change averaged 1.02 points and the FRB/US reading 1.01. The difference was −0.01, because two offsetting movements cancelled: equity valuations fell by 5.04 log points a year, which contributes

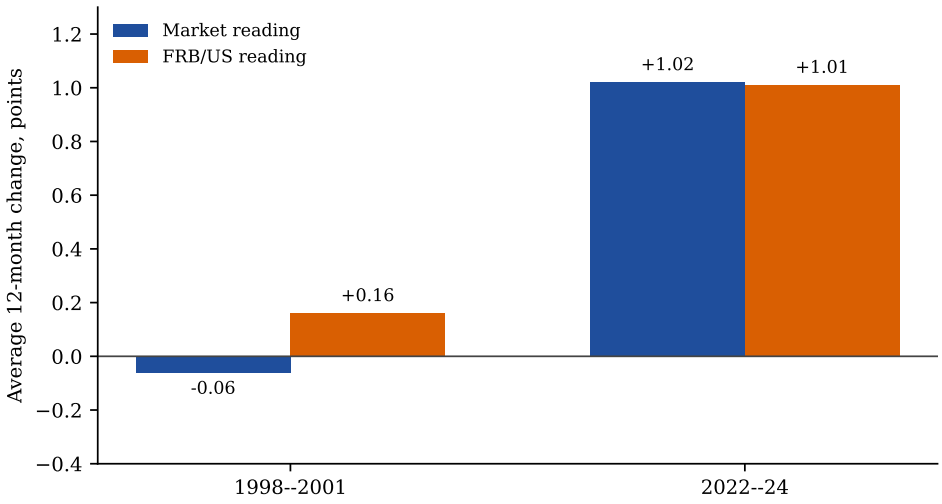


Figure 4: Two Readings of Two Episodes

Notes: Average 12-month change over the months of the episode, in points of the two-year yield (market) and of the funds rate (FRB/US, normalized so that a conventional policy surprise reads one), computed on the same three assets relative to r^*. These are retrospective estimates.

−0.09 to D, and the dollar appreciated by 2.85, which contributes 0.11. The tightening was part equity-led and part dollar-led, and the model's heavier weight on the dollar and lighter weight on equities offset each other. The full indexes differ in average level by 0.96 points (interval −0.05 to 1.79), with the equity weights (0.53) and the dollar (0.52) both contributing, and that difference does not affect the readings of the year's tightening. Unemployment rose by 0.27 points over the following year, against forecasts of 0.65 from the market reading and 0.66 from the FRB/US reading. Both weightings overpredicted the rise, so the weights do not explain the miss.

2004–06. The matched readings are close (the difference is −0.09), and the gap in levels is −0.17 (interval −0.81 to 0.41). The equity weights add 0.65 and house prices subtract 0.73, so the full FRB/US index contains a variable that the market index does not have, and equities do not explain the gap.

These episodes give a concrete answer to when the choice of weights matters. The model gives less weight than the event study to equities and more to the dollar, so the two readings part when valuations and the dollar rise together, as in 1998–2001, and they offset when valuations fall as the dollar rises, as in 2022–24.

6 Can Subsequent Activity Tell the Weightings Apart?

6.1 The Test

Consider two years in which the market index tightened by the same amount. If the dollar accounted for more of the tightening in one year, did activity subsequently weaken more? If the market reading

Table 7: Can Realized Activity Tell the Two Weightings Apart?

	n	Market index β [90%]	Composition γ [90%]	$\gamma - \beta$ [90%]
Unemployment gap, 12 months: 1976–2026	556	0.19 [0.05, 0.33]	−0.27 [−0.96, 0.42]	−0.46 [−1.03, 0.11]
Unemployment gap, 12 months: 1996–2026	318	0.01 [−0.20, 0.22]	−1.18 [−2.51, 0.14]	−1.20 [−2.33, −0.06]
Unemployment gap, 12 months: Excluding 2008–15	436	0.29 [0.15, 0.43]	0.32 [−0.13, 0.77]	0.03 [−0.31, 0.37]
Real GDP growth, four quarters: 1976–2026	186	−0.16 [−0.47, 0.16]	0.89 [−0.77, 2.55]	1.05 [−0.38, 2.47]
Real GDP growth, four quarters: 1996–2026	106	0.31 [−0.13, 0.74]	1.82 [−0.77, 4.40]	1.51 [−0.72, 3.74]
Real GDP growth, four quarters: Excluding 2008–15	146	−0.33 [−0.61, −0.04]	0.27 [−1.20, 1.75]	0.60 [−0.69, 1.89]

Notes: Local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) on own lags and $y = \cdots + \beta M + \gamma D$, where M is the 12-month change of the market index relative to r^* and D the 12-month change of the normalized FRB/US reading less the market reading, computed on the same changes (HAC standard errors, 18 lags, joint covariance of the coefficients). If the market reading is right, $\gamma = 0$. If the normalized FRB/US reading $M + D$ is right, $\gamma = \beta$. A failure to reject is not evidence for either.

is a sufficient summary of the asset prices, the answer is no. If the FRB/US reading is, the answer is yes, by the amount that the model's weights imply. I regress the 12-month-ahead change in the unemployment gap on own lags and

$$y_{t+12} - y_t = \cdots + \beta M_t + \gamma D_t, \tag{3}$$

where M is the 12-month change in the market index (relative to r^*) and D the 12-month change of the normalized FRB/US-minus-market difference on the same changes. If the market reading is sufficient, $\gamma = 0$. If the FRB/US reading $M + D$ is sufficient, $\gamma = \beta$. I read the two restrictions as tests of whether each index summarizes the predictive information in the assets for the outcome specified. That wording matters for unemployment, since the FRB/US weights are constructed from effects on GDP; four-quarter GDP growth is the model-facing check. This is the only primary test, and Table 7 reports both restrictions with the joint covariance of the estimates.

6.2 The Answer

On 1976–2026 the coefficient on the index change is 0.19 and on composition −0.27 (−0.96 to 0.42); the difference $\gamma - \beta$ is −0.46 (−1.03 to 0.11). Neither restriction is rejected. Moving D from its 5th to its 95th percentile at the same index change shifts the year-ahead outlook by −0.30 percentage point, with an interval from −1.06 to 0.47, so the sample cannot establish an outlook difference of either sign. On 1996–2026 the coefficient on composition is −1.18 (−2.51 to 0.14) and the model's restriction is rejected ($\gamma - \beta = -1.20$, −2.33 to −0.06); excluding 2008–15 the coefficients are 0.32 and 0.03, close to the model's restriction. The sign changes across samples. The market restriction is not rejected in any of 12 blocks, and the model's restriction is rejected in 2 of them, without correction for the number of tests. For GDP growth neither restriction is rejected, and adding D does not significantly improve out-of-sample forecasts (Clark–West −1.8).

6.3 Equities Alone

Equity valuation alone raises the adjusted R^2 of the change in unemployment by 0.115, the market index by 0.136, the FRB/US weights by 0.115, the 10-year yield by 0.099, the dollar by 0.004 and the three variables entered freely by 0.147. Both indexes draw heavily on equity valuation, which helps explain why unemployment data have difficulty distinguishing between them: the movements that distinguish the weightings are the ones in which the dollar and equity valuation move in different proportions, and there are few of them. Entered freely, the coefficients are not significantly different from proportional to the market weights ($p = 0.53$) or to the FRB/US weights ($p = 0.23$); the power is low.

6.4 What Forecasters Expected and What Happened

Forecasters' errors combine the outcome with what they had already anticipated. Table 8 regresses the Survey of Professional Forecasters' unemployment forecast, the realized change and the error (realized less forecast), over the same 128 surveys (1990Q3–2025Q3), on the index change and D. The coefficients on the forecast and the error add to the coefficient on the realization. When FRB/US reads the same asset movements as tighter than the market index does, the forecast does not respond (-0.18, interval -0.61 to 0.25), unemployment tends to come in lower (-1.47, -2.84 to -0.10), and so the error is negative (-1.29, -2.38 to -0.21). Composition predicts what subsequently happened, not what forecasters expected, and with a sign opposite to the one the FRB/US reading would imply if forecasters had not priced composition. The result depends on the sample. Excluding 2008–15 the coefficients are 0.29 for the forecast and -0.17 for the realization; the error keeps its sign (-0.46, interval -0.89 to -0.04) and its interval now includes zero. The outcomes overlap, the coefficients are not adjusted for their number, and 2001 and 2008–09 probably dominate.

What can a policymaker take from this? The two readings can be used interchangeably when markets move as they do around policy news, and the data cannot say which to prefer when they do not. A reading from either index should be accompanied by the question of what moved: the dollar, valuations, or yields. The answer indicates whether the two readings are likely to part.

Table 8: What the Composition Predicts: Forecasts, Outcomes and Errors

	n	Coefficient on D [90%]	Effect, 5th to 95th percentile
Unemployment, 1990–2025: Survey forecast	128	−0.18 [−0.61, 0.25]	−0.16
Unemployment, 1990–2025: Realized	128	−1.47 [−2.84, −0.10]	−1.33
Unemployment, 1990–2025: Forecast error	128	−1.29 [−2.38, −0.21]	−1.16
Unemployment, excluding 2008–15: Survey forecast	88	0.29 [−0.06, 0.65]	0.25
Unemployment, excluding 2008–15: Realized	88	−0.17 [−0.84, 0.49]	−0.15
Unemployment, excluding 2008–15: Forecast error	88	−0.46 [−0.89, −0.04]	−0.39
Real GDP growth, 1990–2025: Survey forecast	128	−0.35 [−0.93, 0.23]	−0.31
Real GDP growth, 1990–2025: Realized	128	2.17 [−0.05, 4.38]	1.95
Real GDP growth, 1990–2025: Forecast error	128	2.52 [0.49, 4.54]	2.27

Notes: Regressions of the Survey of Professional Forecasters' median forecast of the four-quarter-ahead change in the unemployment rate (real GDP growth), the realized change, and the forecast error (realized less forecast) on the 12-month changes of the market index and of D, financial data up to the month before each survey's deadline, 1990Q3–2025Q3; the coefficients on the forecast and the error add to the coefficient on the realization. HAC standard errors; effect in percentage points at the same index change. Incremental information only: the test says nothing about causation.

7 Conclusion

The paper asks when a movement in financial conditions can be read as a movement in monetary policy under two calibrations. For the usual response to a policy surprise, the two readings agree: the model prices it at 0.975 funds-rate points with persistent paths and 0.83 with a half-life of four quarters, and after rescaling they agree exactly along that direction. For other combinations of asset prices the readings differ, because the model gives the dollar about 2.49 times the relative weight that the event study does and common assumptions about decay do not alter the ratio. The difference is a quarter to a third of the average market move at horizons from a day to a year.

Realized activity does not separate the two. Conditional on the market index, composition does not predict unemployment or GDP with a sign or size that the sample can pin down, and equity valuation alone carries most of the content. These are statements about measurement. The causal effect of a movement on spending and its source remain separate questions; a movement that departs from the policy-news mix can come from the announcement itself (Boehm and Kroner, 2025).

The limits are those of the design. The FRB/US weights come from one model and one experiment and are held fixed in the intervals; the market weights are imprecise; the horizons are compared on overlapping data; the daily comparison uses nominal assets; and there are five recessions in the sample. The most durable finding is the one the episodes show. The choice of weights moves the reading when valuations and the dollar rise together: in 1998–2001 the two readings of the same asset movements had opposite signs. It does little when they offset, as in 2022–24, when the two readings of the year's tightening were the same and both overpredicted the rise in unemployment. A reader who wants to use either index can now say when to expect them to differ.

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Appendix A The Weights: a Derivation

Let the movements in the three asset prices be $x = \psi s + u$, where s is policy news, ψ the responses and u everything else, with covariance Σ. Let g_h be the effects of a persistent unit shift in each asset price on spending at horizon h. The FRB/US weights are proportional to g_h. The market weights solve the problem of recovering s from x: the generalized least squares estimator is $\hat{s} = w_M' x$ with $w_M = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$, which satisfies $w_M'\psi = 1$ and, by the Gauss–Markov theorem, minimizes the variance of the estimate among linear unbiased estimators of s when u has mean zero and covariance Σ and is uncorrelated with s. For any x the difference between the normalized readings is $D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u$, because the policy part cancels. The readings coincide for every movement if and only if $w_{B,h} = w_M$, that is, if $g_h \propto \Sigma^{-1}\psi$: the criterion that recovers policy news precisely and the criterion that measures spending effects select the same weights only in that case. The variance of the difference is $(w_{B,h} - w_M)'\Sigma(w_{B,h} - w_M)$.

Observed movements need not be of the form $\psi s + u$ with a single policy factor, and the model's effects need not be those of a movement of the same shape. If all observed movements lay along the single direction ψ, every weight vector normalized to price ψ at one would give the same reading on them. The data would then identify the reading along that direction and leave the weights themselves undetermined. In the announcement windows u is not zero, and it is the variation in u that identifies Σ and hence the weights. What distinguishes the two weightings in practice is the exposure of the difference in weights to the movements that occur, which Table A.1 measures as a ratio to the market move.

Appendix B Construction of the Series

The three asset prices enter in three forms, each answering a problem of the one before. The headline results use the third.

Nominal. Rates as published. A nominal index treats a 10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable,

Table A.1: How Large is the Difference Between the Two Readings?

Movements	Mean $\lvert D\rvert$	Ratio to market move [90%]	Share above 0.25 (%)	Mean run (months)
Announcement windows, held out	0.010	0.19 [0.09, 0.36]	0	–
Monthly changes, 1996–2026	0.042	0.24 [0.14, 0.53]	0	1.0
Monthly changes, 1973–2026	0.062	0.26 [0.15, 0.52]	2	1.1
12-month changes, 1996–2026	0.25	0.33 [0.16, 0.66]	36	7.4
12-month changes, 1973–2026	0.32	0.31 [0.16, 0.53]	44	7.0

Notes: D is the FRB/US reading (four quarters, normalized so that a conventional policy surprise reads one) less the market reading, applied to the changes of the equity valuation, the 10-year yield and the dollar at the r^* step. Announcement windows: held-out events (weights re-estimated without each event). Ratio: mean $\lvert D\rvert$ over mean absolute market move; interval from bootstrap samples of the announcements and, for the monthly history, of 12- or 24-month blocks of months. A run is consecutive months with $\lvert D\rvert$ above 0.25.

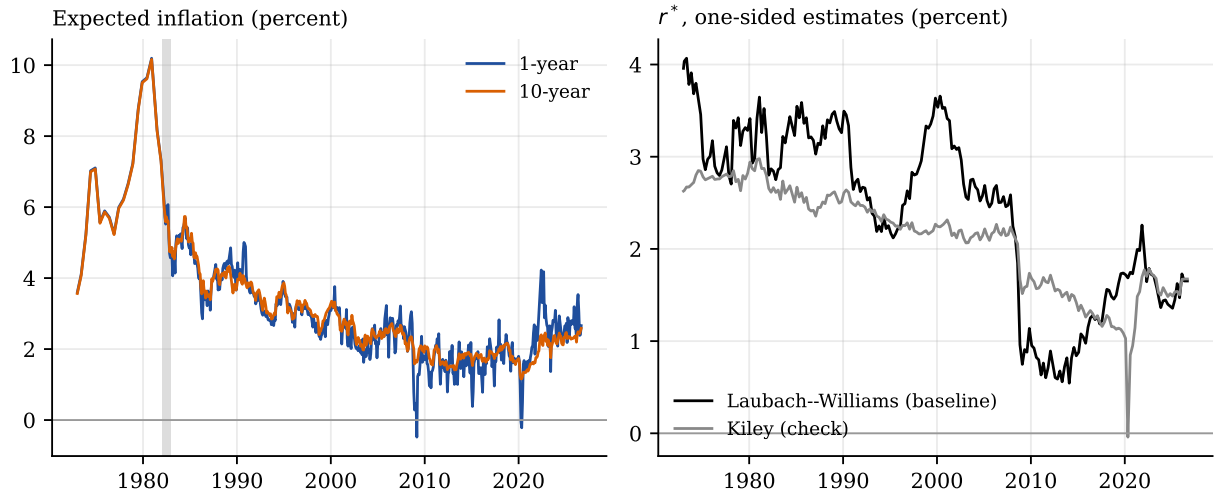


Figure B.1: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure B.1).

Real. Each rate less expected inflation at a matched horizon: ten-year expected inflation for the 10-year yield. The dollar is the real broad index.

Relative to r^*. The real 10-year yield less r^*. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$, which is also an implicit real rate, as the gap $E/P - r^*$, converted to CAPE log points at the sample-average earnings yield so that its weight on CAPE is unchanged. The baseline uses the one-sided, full-sample Laubach–Williams estimate. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

The transformation matters for levels and not for the difference between the two readings. The ratio of the mean absolute difference to the mean absolute market move in 12-month changes, 1996–2026, is 0.33 (interval 0.18 to 0.69) with nominal assets and 0.33 relative to r^*.

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table B.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad

dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4 in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time; draws use the seed in `lib/config.py`.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. For the decay experiments of Section 3.3, each variable's shock decays geometrically with the half-life stated, and the response is computed in the model by linearity from the temporary-shock responses.

Forecasting and survey data. The regressions use local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) with HAC standard errors of 18 lags. The Survey of Professional Forecasters data are the median forecasts, 1990Q3–2025Q3, matched to the financial data up to the month before each survey's deadline.

Appendix C Inference and the Decomposition of the Levels

C.1 How Precisely Are the Weights Estimated?

The weights come from regressions on 253 announcements. The weight on the 10-year yield has a 90 percent interval from 0.13 to 1.13 around 0.61. The interval for equity valuation runs from -0.07 to -0.03, and for the dollar from 0.02 to 0.11. The FRB/US weights on all three lie inside these intervals, the equity weight near the edge: it is lower in absolute value than 7 percent of the bootstrap draws, about 14 percent in a two-sided test. The data do not reject that the model weights equal the market weights. This is a statement about power, not a finding that they are equal, and it treats the FRB/US weights as fixed. The intervals in the text come from the bootstrap of the announcements; the contrast in composition in Section 3.3 is the ratio of ratios, whose interval includes one at two quarters.

C.2 From the Market Index to the FRB/US Index

Table C.1 accounts for the difference in average levels between the full FRB/US index and the market index, in stages on the same inputs and centering: the variables the market index does not

Table B.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment; real GDP	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU; GDPC1
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
Survey of Professional Forecasters	Quarterly	Federal Reserve Bank of Philadelphia
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

have (the policy rate, spreads and house prices), the equity weights, the dollar weights and the 10-year weights.

Appendix D Sensitivity

D.1 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside at 10 dates in 2011–14, by at most -0.073 against a band edge of -0.070 (Figure D.1). A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005. The 10-year weight is imprecise throughout. Apart from the equity weight in 2011–14, the weights stay within their sampling error.

D.2 Other Policy-News Series and Event Samples

Table D.1 re-estimates the weights with the two-year yield as the policy news, with announcements through 2019, and without the lower-bound years. The equity weight is -0.027, -0.061 and -0.039

Table C.1: From the Market Index to the FRB/US Index: Shares of the Difference

Stage	Level, share of variance (%)	12-month change (%)
Equity weights	72 [-1, 91]	49 [-6, 87]
Dollar weights	14 [-1, 38]	18 [-1, 35]
10-year weight	1 [-4, 53]	2 [-7, 57]
Normalization of the weights	3 [-5, 15]	2 [-1, 13]
Policy rate	-0 [-1, 1]	0 [-1, 1]
BBB spread	1 [-1, 8]	4 [-2, 16]
Mortgage spread	3 [-3, 15]	17 [-0, 44]
House prices	8 [-2, 34]	9 [0, 24]

Notes: Shares of the variance of the FRB/US index (with house prices) less the market index, relative to r^*, 1973–2026, attributed to each stage of a bridge between them: the weights on the three common assets, the normalization of the weights, then the contributions that only the FRB/US index has (policy rate, spreads, house prices); both indexes use the same inputs and centering. Intervals (5th to 95th percentile) from bootstrap samples of the announcements and of blocks of months.

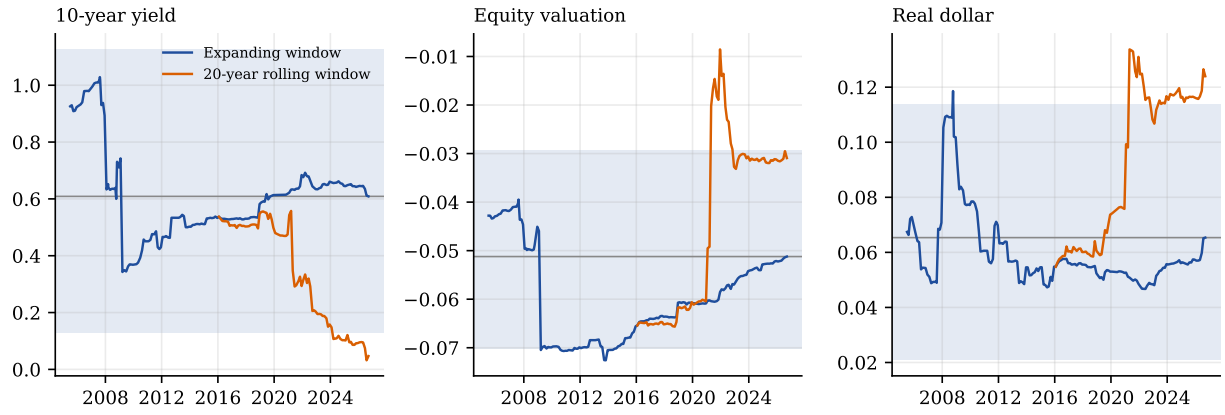


Figure D.1: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.

Table D.1: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69

Notes: Weights w and responses ψ as in Table 2, re-estimated on each sample or with the window change of the two-year yield as the policy-news measure. In each case the weights applied to the responses sum to 1.

against −0.051 in the baseline, and the conclusions about composition rest on the model's ratio of dollar to equity effects, not on any one set of market weights.

D.3 Limits of the Design

The weights come from announcements since 1996 and are applied from 1973; they assume that the responses reflect policy rather than information effects (Nakamura and Steinsson, 2018), and if policy came to target financial conditions the responses could change, the Lucas-critique concern that Hatzius et al. (2010) raise for all indexes. The FRB/US weights come from one model, at one horizon and in one experiment, and are held fixed in the intervals. The daily comparison of Section 4 uses nominal assets, with the nominal broad dollar for the real dollar, no adjustment for r^* and weights fitted on the same announcements; the intraday comparison holds each event out. The sample has five recessions, two of them out of sample, and the 12-month outcomes overlap, so the counts of tests are descriptive and not adjusted for their number. The neutral rate is not neutral to outcomes, because the Laubach–Williams model uses output, inflation and unemployment.